

ECON 100B: Macroeconomics

Class 9: Growth and Ideas 1 (Jones Chapter 6)

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February 17, 2026

Agenda

- Announcements
- Current events
- A look at the math of the steady state in Chapter 5
- Chapter 6: Growth and Ideas



BERKELEY STUDENT LEARNING CENTER

ECONOMICS PROGRAM

Mastery through Collaborative Learning

ECON 100B

César Chávez Student Center

Monday - Thursday 12 - 5 PM



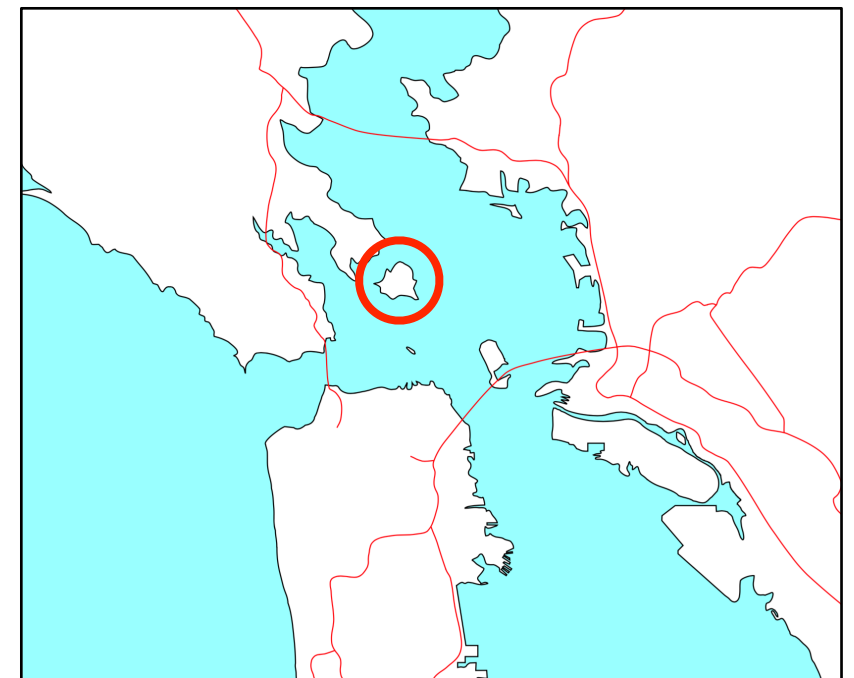
Service Schedule

Announcements

- We'll do some Solow math today. The “Golden Rule” material at the end of Class 08 is *not something you must know*, but it could appear verbatim on the midterm in a guided fashion
- Credo: Everything in the slides is either a learning objective or an example. We'll cover learning objectives.
- The “Golden Rule” is an example with some calculus. Calculus is not on the exam. This is ECON 100B, not 101B.
- Problem Set 2 will be due on Gradescope on Friday 2/27
- In-class Midterm the next Thursday 3/5
- Take-home Midterm Essay released 3/8, due 3/20
- Spring Break 3/23-3/27

Local SF Bay History

- Traditionally, immigration is associated with Ellis Island in New York harbor near Statue of Liberty
- Here in the SF Bay, we also have our own “Ellis Island”
- *United States Immigration Station on Angel Island*, AIISF
- You can take a ferry out there from SF or Tiburon
- Self-guided and guided tours are available
- Large amount of Chinese and other poetry written on the walls
- It’s a moving experience



AIISF was many things, including unauthorized

- [Homer Lee \(1908-2009\)](#) was a Chinese immigrant and resident of Berkeley. In this [2003 article on SFGate](#), Homer was candid about the circumstances
- Like his older brother, he was a “[paper son](#)” — family connection (forged) on paper
- He sought entrance to the U.S. in 1926, after the 1906 earthquake and fire destroyed paper records in San Francisco and his father had migrated in 1912
- The end of China’s Qing Dynasty in 1912 was turbulent
- Homer Lee was an unauthorized immigrant. He was kept at AIISF for 6 months

“In 1966, during an amnesty period, Homer Lee admitted he had illegally identified himself as the son of a citizen when he arrived in 1926. He applied for naturalized citizenship, studied for the exam and in 1969 was sworn in as a naturalized citizen.” ([St. John, 2003](#))

Read both of these newspapers

- Wall Street Journal
- New York Times
- For access to both, see [this Library guide](#)
- You might skip the op-ed pages unless you like being horrified or vindicated, depending on your point of view
- But sometimes there are very good op-ed pieces in both

Current events

- Nick Timiraos (WSJ, Feb 14, 2026) “[The Economy May Have Stuck the Soft Landing. Nobody Wants to Jinx It.](#)”
 - A “soft landing” means calm after a period when inflation rose sharply
 - As we’ll see in the second half of ECON 100B, there have been many “hard landings,” with very high unemployment after high inflation and after the Fed’s reaction to high inflation
- Ruth Simon (WSJ, Feb 15, 2026) “[The Break Is Over. Companies Are Jacking Up Prices Again.](#)”

Current events in economic growth

- Rebecca Robbins (NYT, Feb 16, 2026) “[Vaccine Makers Curtail Research and Cut Jobs](#)”
- Moderna produced one of the two mRNA vaccines for Covid-19
- Last week, FDA decided not to review Moderna’s new mRNA-based vaccine against the seasonal flu
 - Hometown hero Vinay Prasad of UCSF spearheaded this move
 - Prasad argues Moderna ran their international trial with a control group that was not business-as-usual: not an extra dose for age 65+
 - Moderna replies that FDA had not highlighted that beforehand
 - [WSJ editorial board is not pleased](#). Neither are Moderna shareholders

First half of ECON 100B

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
 - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y]$$

$$g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

- Chapter 4 introduces Cobb-Douglas production

$$Y = A K^{1/3} L^{2/3}$$

with constant returns to scale
and declining marginal productivity of K or L

- Chapter 5 develops the Solow Model
- **Chapter 6 develops the Romer Model**

Back to Slides 28-31 in Class 08

“Math is power”

-Every economist who ever lived

[Per the NYT](#), it's also a tag line of 2024 documentary film “[Counted Out](#)”

Solving for the steady state

$$\Delta k = sy - (n + d)k$$

write out y

$$\Delta k = sAk^{1/3} - (n + d)k$$

When the economy is in steady state, $\Delta k = 0$, so therefore

$$\Delta k = sAk^{1/3} - (n + d)k = 0$$

$$sAk^{1/3} = (n + d)k$$

Collect terms

$$\frac{sA}{n + d} = \frac{k}{k^{1/3}} \quad \rightarrow \quad \frac{sA}{n + d} = k^{2/3}$$

Raise to 3/2

$$k^* = \left(\frac{sA}{n + d} \right)^{3/2}$$

Steady-state output

$$k^* = \left(\frac{sA}{n+d} \right)^{3/2}$$

$$y^* = Ak^{*1/3} \quad \rightarrow \quad y^* = A \left[\left(\frac{sA}{n+d} \right)^{3/2} \right]^{1/3}$$

Simplify

$$y^* = A \cdot A^{3/2 \cdot 1/3} \left[\left(\frac{s}{n+d} \right)^{3/2} \right]^{1/3}$$

$$y^* = A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}$$

The capital-output ratio

$$k^* = \left(\frac{sA}{n+d} \right)^{3/2} \qquad y^* = A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}$$

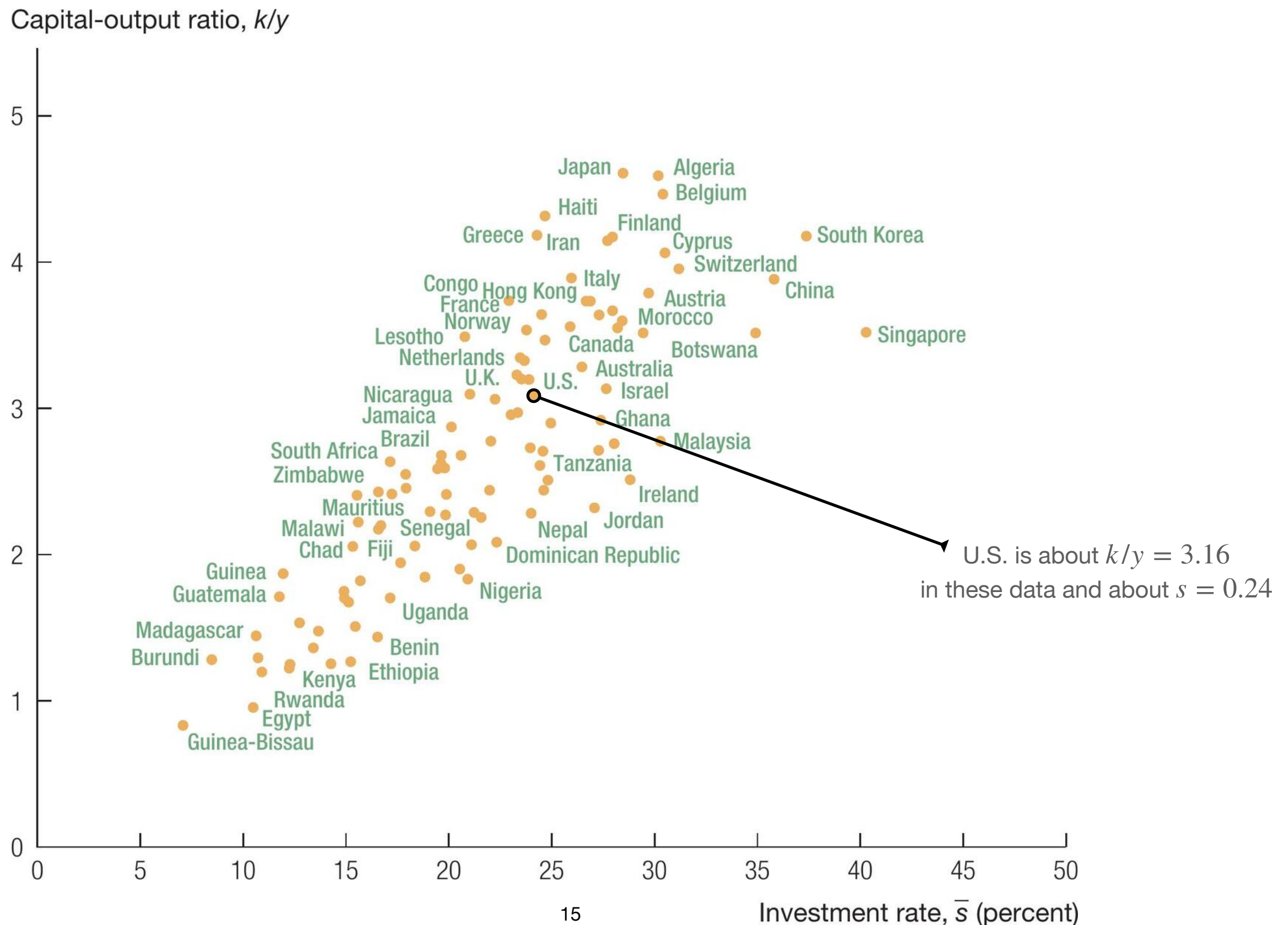
- Note that $\frac{K}{Y} = \frac{k}{y}$ because $k = \frac{K}{L}$ and $y = \frac{Y}{L}$

$$\frac{k^*}{y^*} = \frac{\left(\frac{sA}{n+d} \right)^{3/2}}{A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}} \qquad \frac{k^*}{y^*} = \frac{A^{3/2} \left(\frac{s}{n+d} \right)^{3/2}}{A^{3/2} \left(\frac{s}{n+d} \right)^{1/2}}$$

$$\frac{k^*}{y^*} = \frac{s}{n+d}$$

$$\frac{k^*}{y^*} = \frac{0.24}{0.01 + 0.05} = 4$$

Figure 5.3 in §5.5 shows $\frac{k}{y}$ as a function of s across countries



Learning objectives in Chapter 6

- Goods and services are produced by **ideas**, capital, and labor
- Ideas are special
 - They are **nonrival** and can be used in many places at the same time
 - New, useful ideas might be scarce, but existing ideas are not scarce
 - Ideas can still be **excludable**, legally kept from use by others
- If ideas can also be produced, they help us achieve increasing returns to scale and growth in income per person
- To produce ideas, we probably need
 - An **ideas sector** with labor
 - Some kind of tool to make ideas excludable, like patent and copyright law

The “old school” Romer model

- In ECON 100B, we will learn and use the Romer model appearing in 1e through 5e of Chad Jones’s *Macroeconomics*
- In 6e, a different version appears in §6.3
 - It’s a little harder
 - To me, its benefits don’t seem to outweigh its costs
- Interested students are welcome to look at the new version

Let's rethink how we model the world

- Up until now, we have focused exclusively on physical **objects** — capital (machinery, plants, tools), people, and much of Gross Domestic Product are indeed physical objects
- This wasn't a mistake, but we have seen a chief limitation:
- We still don't have a theory of long-run growth in income per person
- But physical objects are not the only things in the world that either have or can create value
- We already know that services have value, since they are in GDP, even though they are not physical objects ...

Ideas are another input to production

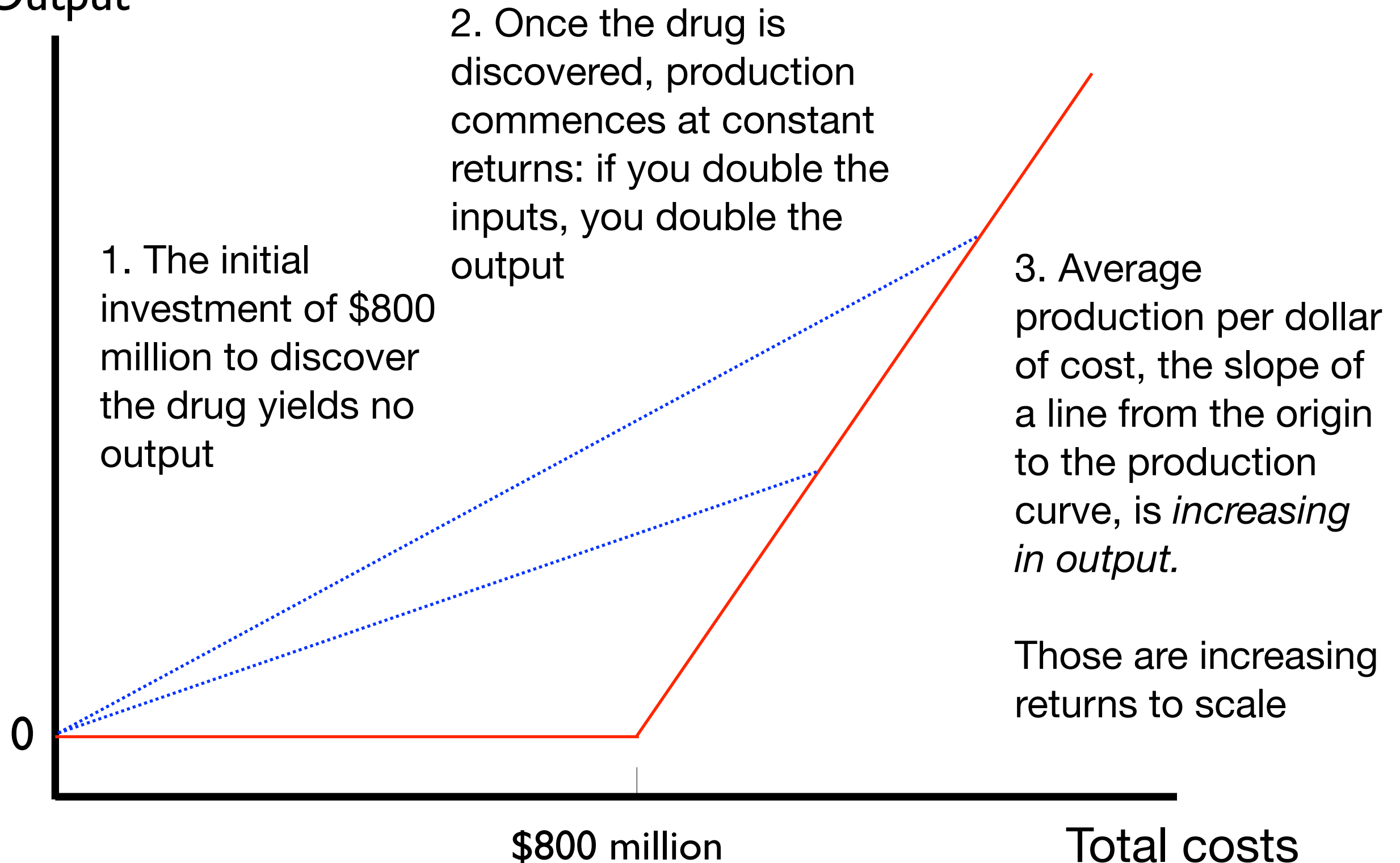
- And ideas are **special**
- Ideas are nonrival: my use doesn't preclude your use, vice-versa
- Ideas produce increasing returns because of their nonrivalry: Once you've spent enough to develop a valuable idea, you can use it over and over in many places at once
- This suggests that ideas may help us escape the stagnation of constant returns and actually explain the sources of growth
- But competitive markets may *underproduce* ideas: because there are high fixed costs and increasing returns only later

A real-world example helps explain increasing returns

- Consider a pharmaceutical company
- It has workers (L) and capital equipment (K) like labs, offices, plants for producing drugs
- But the most important ingredients are ideas — about how to combine molecules into drugs (New Molecular or Chemical Entities, NMEs/NCEs)
- Coming up with ideas is expensive. Developing a new drug costs something like \$800 million in 2005 dollars
- But once you have the drug, you can produce copies fairly cheaply using capital and labor and then sell all those copies

How does this become increasing returns?

Output



Why do we see increasing returns?

- The nonrivalry of the **drug idea** means that once you spend enough to discover it, you can use it all you want without depleting it
- This is definitely not the same as when you buy capital goods like plants or equipment or when you employ labor!
- You can't employ someone in two places at once, but you can use an idea anywhere at any time
- Since there are constant returns using capital and labor alone, adding in ideas makes total returns *increasing*

What are the consequences of increasing returns?

- Competitive markets may **underproduce new ideas** and anything in which there are increasing returns
- Why?
- Under perfect competition, firms produce where price equals marginal cost
- But the marginal cost of producing a unit of a drug is tiny, even if the cost of developing it was very large
- If competition pushes prices of drugs down to their marginal costs immediately, then nobody would ever develop a drug for fear of losing the fixed costs of development

With pharmaceuticals and medical treatments, there is an additional reason why markets underproduce

- Some medical conditions are common; other medical conditions are extremely rare
- The science can be difficult for either type
- [Malaria affects hundreds of millions](#) of people worldwide
 - The Gates Foundation provided most of the funding to develop [RTS,S \(Mosquirix\)](#), approved finally in the early 2020s
 - The science was hard and most demanders are poor
- [Sickle-cell disease](#) is relatively rare
 - [Affects 100,000 Americans](#), or about 0.03%
 - [New gene therapies in late 2023](#) based on CRISPR, from [Berkeley Nobel Laureate Jennifer A. Doudna](#) & colleagues: \$3 million per patient to cure

How do we get around this problem?

- **Patent and copyright law** produce legal but time-limited monopolies which can charge higher prices and recoup costs
- **Grants and philanthropy**
 - Government research and development (R&D) funds are publicly provided using taxpayer money
 - National Science Foundation \$9 billion annual budget
 - National Institutes of Health \$48 billion annual budget
 - DARPA (Defense Department) \$4 billion annual budget
 - The wealthiest people (and posthumously, their charitable organizations) frequently donate, run contests with prizes, etc.
 - Gates Foundation, from Microsoft — [\\$9 billion per year until 2045](#)
 - John D. and Catherine T. MacArthur Foundation, wealthy from the insurance industry — perhaps \$0.5 billion per year

The economics of ideas is characterized by increasing returns to scale

- Previously, our production model has exhibited constant returns to scale in physical inputs —
- If you duplicate a factory with all its workers (L) and machinery (K), you should also duplicate output
- This is still a very good model of reality!
- But let's think about formally treating ideas as another input
- We already know that if we double capital (K) and labor (L), duplicating the factory, we get double the output (Y)
- What if we also double the amount of ideas?

Increasing returns in the production function when A , K , and L all scale up

Our production function is

$$Y = F(A, K, L) = AK^{1/3}L^{2/3}$$

If we duplicate all inputs, and not just the physical inputs K and L , we find that

$$F(2A, 2K, 2L) = (2A)(2K)^{1/3}(2L)^{2/3}$$

$$F(2A, 2K, 2L) = 2 A 2^{1/3} K^{1/3} 2^{2/3} L^{2/3}$$

$$F(2A, 2K, 2L) = 2 2^{1/3} 2^{2/3} AK^{1/3}L^{2/3}$$

$$F(2A, 2K, 2L) = 2^2 AK^{1/3}L^{2/3}$$

$$F(2A, 2K, 2L) = 4 F(A, K, L) \quad \text{increasing returns!}$$

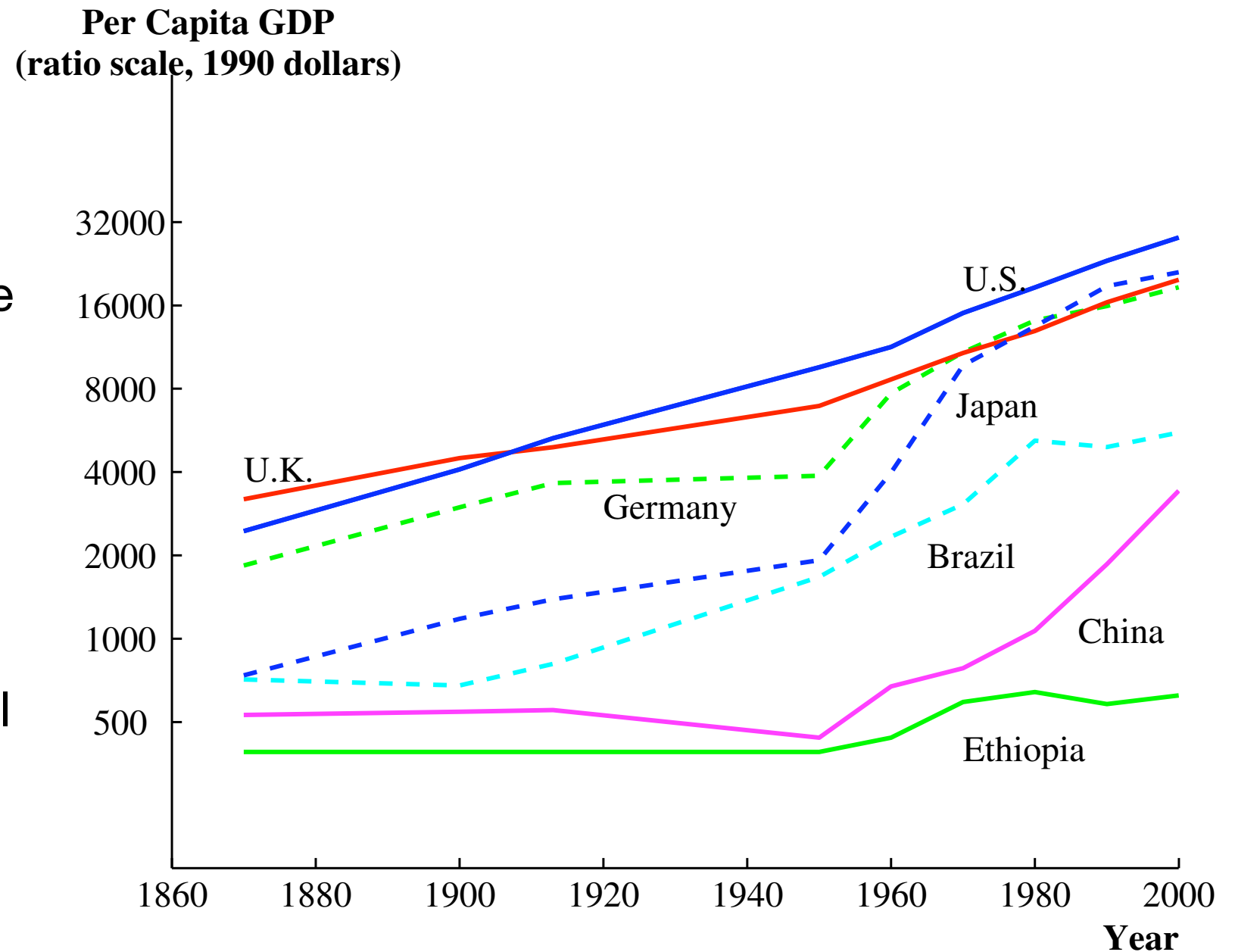
Why might increasing returns be useful when studying growth?

So far we have seen what capital and labor **can't explain**:

1. Sustained growth over time
2. Big differences between rich and poor countries

This is because of the **diminishing returns** to capital and labor ...

So maybe there's something that doesn't have diminishing returns ...



Note: Data from Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003). Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

The Romer model of growth and ideas

- Like before, suppose there is one production good **Y**
- Let **A** be the stock of ideas, and new ideas are produced just like Y
- For simplicity, we'll forget about capital and imagine that **labor** is the only factor of production
- Labor is rival — people can either work in goods (Y) production OR in idea production (A) but not both
- But ideas are **nonrival** — ideas can work in goods (Y) production and in idea production (A) at the same time

An aside: Stocks of factor inputs produce flows of production through production functions

- The output of a production function is a flow of something new, like new cars or new potato chips
- Production functions take as inputs some stocks of productive factors, like technology, capital, and labor
- Both of these production functions produce a flow as a function of stocks

$$Y = F(A, K, L) = AK^{1/3}L^{2/3}$$

a function of capital & labor

$$\Delta L = \Phi(M, D)$$

a “fertility function” of
moms & dads

- In both cases, the factor inputs are stock variables and the outputs are flow variables

Setup of the “old school” Romer model from Jones 1e-5e

- In the goods sector, “food” is a function of ideas and farmer labor L_{yt} . There is no capital, and food rises linearly with farmers

$$Y_t = A_t L_{yt}$$

- In the ideas sector, **new ideas** ΔA_t are a function of current ideas and scientist labor L_{at} . There is no capital, and new ideas rise linearly with scientists, whose productivity rises with $\bar{z}$

$$\Delta A_t = \bar{z} A_t L_{at}$$

- Total labor $\bar{N}$ is split between farm and science. Science share is ℓ

$$L_{at} + L_{yt} = \bar{N} \quad L_{at} = \ell \bar{N}$$

Remarks

$$Y_t = A_t L_{yt}$$

$$\Delta A_t = \bar{z} A_t L_{at}$$

- Ideas are very important, but they also a **means to an end**. The end is utility from consumption, and we can't consume ideas
- Income or “food” per person equals $y_t \equiv \frac{Y_t}{N}$
- More ideas help the economy produce more food, but they come at a resource cost of L_{at} workers who could otherwise be farming
- The magic is that ideas A_t can be used simultaneously in food production and in new idea production. Ideas are nonrival inputs

Solving the model, step 1

- What does our model say about income per person, which is now $Y/\bar{N}$ in this framework?

$$Y_t = A_t L_{yt} \quad \text{and} \quad L_{yt} = (1 - \ell)\bar{N}$$

$$Y_t = A_t (1 - \ell)\bar{N}$$

$$\frac{Y}{\bar{N}} \equiv y_t = A_t (1 - \ell)$$

- Income per person is a function of the total stock of knowledge
- A new idea increases everyone's income because it's nonrival
- By contrast, in the Solow model, income per person depends on capital per person, not the level of capital overall
- This is because capital is a rival good in production. Two workers cannot use the same capital good at the same time

Solving the model, step 2

- Ideas are important. What can we say about growth in ideas?

$$\Delta A_t = \bar{z} A_t L_{at} \quad \text{and} \quad L_{at} = \ell \bar{N}$$

$$\Delta A_t = \bar{z} A_t \ell \bar{N} \quad \text{now divide by } A_t$$

$$\frac{\Delta A_t}{A_t} = \bar{z} \ell \bar{N}$$

- The growth rate of the stock of ideas, $g[A]$, is constant over time and proportional to the number of idea producers in the economy
- Since the growth rate of A is constant at $g[A] = \bar{z} \ell \bar{N}$, we can write:

$$A_t = A_0 (1 + g[A])^t$$

where A_0 is the initial level of ideas

Solving the model, step 3

$$A_t = A_0 (1 + g[A])^t$$

- If this is a formula for the stock of ideas A_t then recall

$$\frac{Y}{\bar{N}} = y_t = A_t (1 - \ell) \quad \text{substitute for } A_t$$

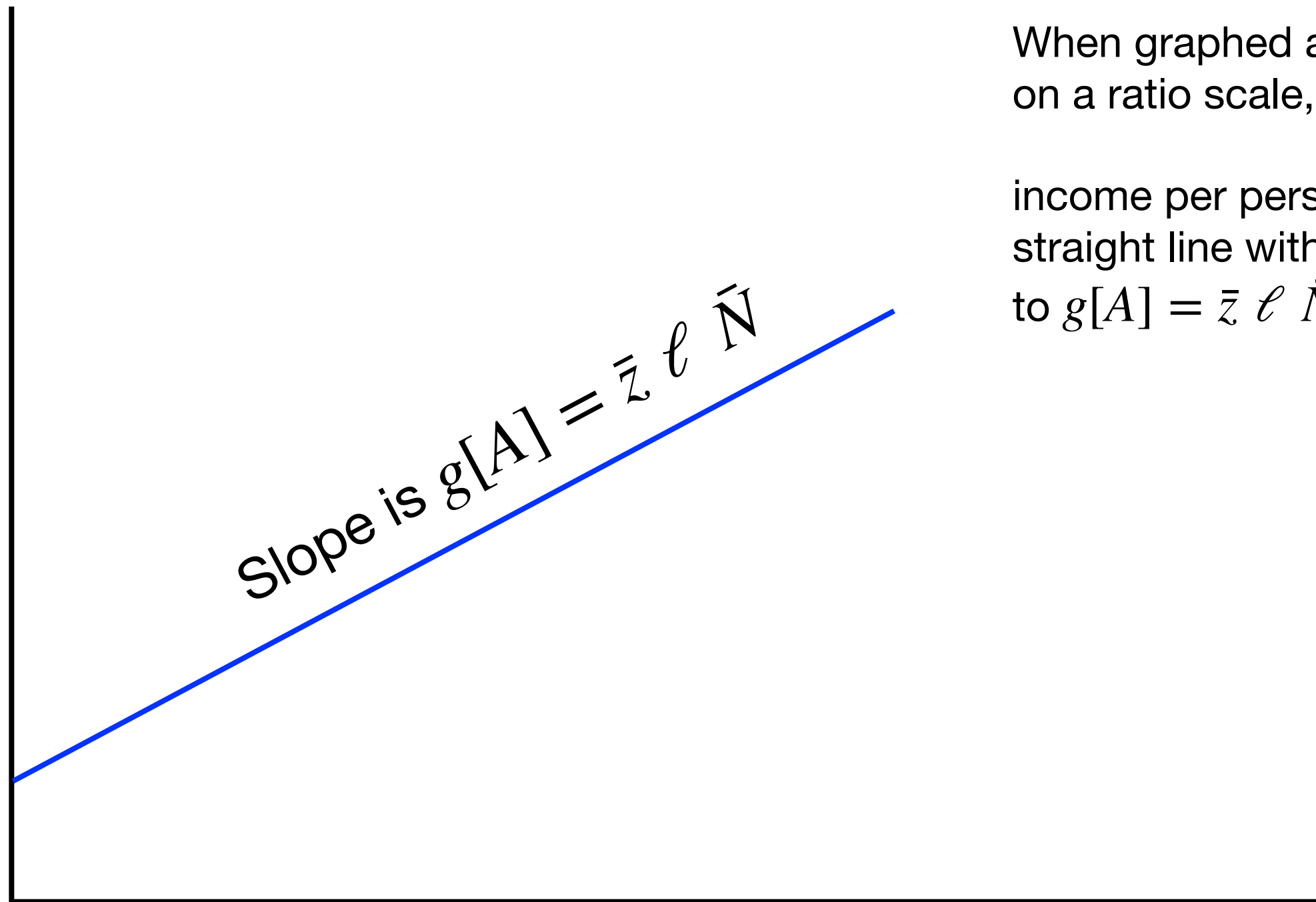
$$y_t = A_0 (1 + g[A])^t (1 - \ell) \quad \text{or rearranging,}$$

$$y_t = A_0 (1 - \ell) (1 + g[A])^t$$

- This is exponential growth in income per capita
- No other model has produced sustained growth over time
- Here the “steady state” is really a “**balanced growth path**”

$$y_t = A_0 (1 - \ell) (1 + g[A])^t$$

Ratio scale: y_t



When graphed against time
on a ratio scale,

income per person is a
straight line with slope equal
to $g[A] = \bar{z} \ell \bar{N}$

Why is there growth in the Romer Model?

- Here, ideas are the engine — in the Solow model, it was capital
- Consider the idea accumulation equation:

$$\Delta A_t = \bar{z} A_t L_{at}$$

- There are constant returns to ideas in idea production:
 - The exponent on A_t equals one
 - As we gain knowledge, the return to knowledge does not fall
- Capital accumulation in the Solow Model is different!

$$\Delta K = s A K^{1/3} L^{2/3} - dK$$

- There are diminishing returns to capital in capital production, because $F(A, K, L)$ exhibits diminishing returns in capital
- Why the difference? Ideas are nonrival, capital is rival

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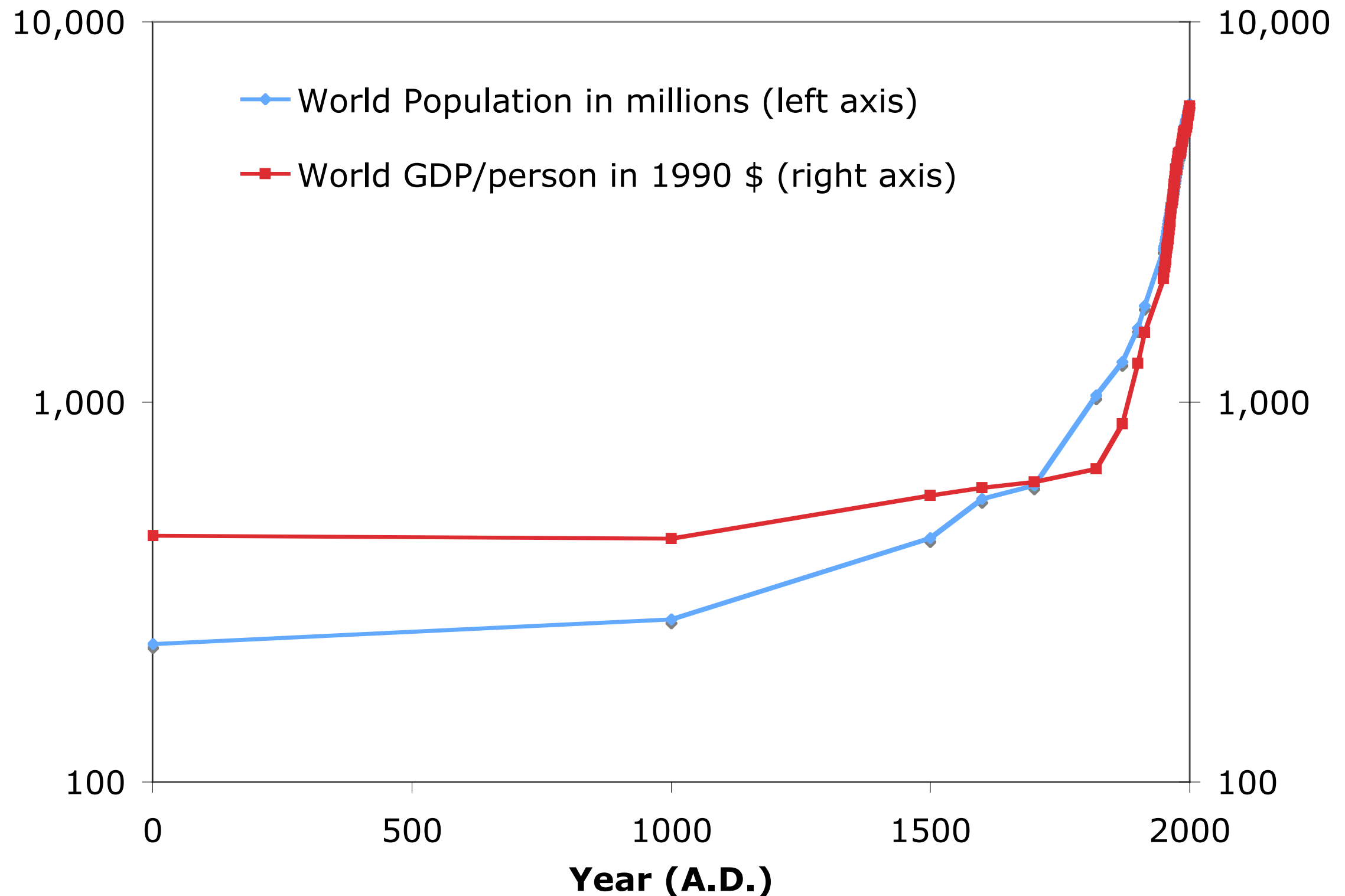
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How does the Romer Model apply to reality?

- One of the central results is that the growth rate in income per capita is the same as the growth rate in ideas, a linear function of population size
- But are growth rates higher in bigger countries?
 - “Tiny” Luxembourg has fewer than half a million people
 - The U.S. has almost 300 million, more than half a million scientists!
 - Between 1960 and 2006, annual growth in Luxembourg was 3.3%, about a third faster than the 2.5% clip enjoyed by U.S. citizens
- What’s going on? Ideas are so nonrival that many countries can benefit from them today
- The Romer model best describes world growth...

Growth in world population preceded and may have caused growth in GDP per capita during the last 2 millennia



What else affects growth in the Romer Model?

- In this model, there are 4 parameters:
 - $\bar{N}$ the size of the population
 - ℓ the share of the population in “science” or ideas
 - $\bar{z}$ the productivity of scientists
 - A_0 the starting level of ideas
- What happens when we change these parameters?
- In the “old school” Romer Model, there is a balanced growth path and no transition dynamics

Suppose $\bar{N}$ increases, what happens to growth in y_t ?

Ratio scale: y_t

$$y_t = A_0 (1 - \ell) \underbrace{\left(1 + g[A]\right)^t}$$

$$g[A] = \bar{z}\bar{\ell}\bar{N}$$

Slope = $\bar{g}$

Time, t

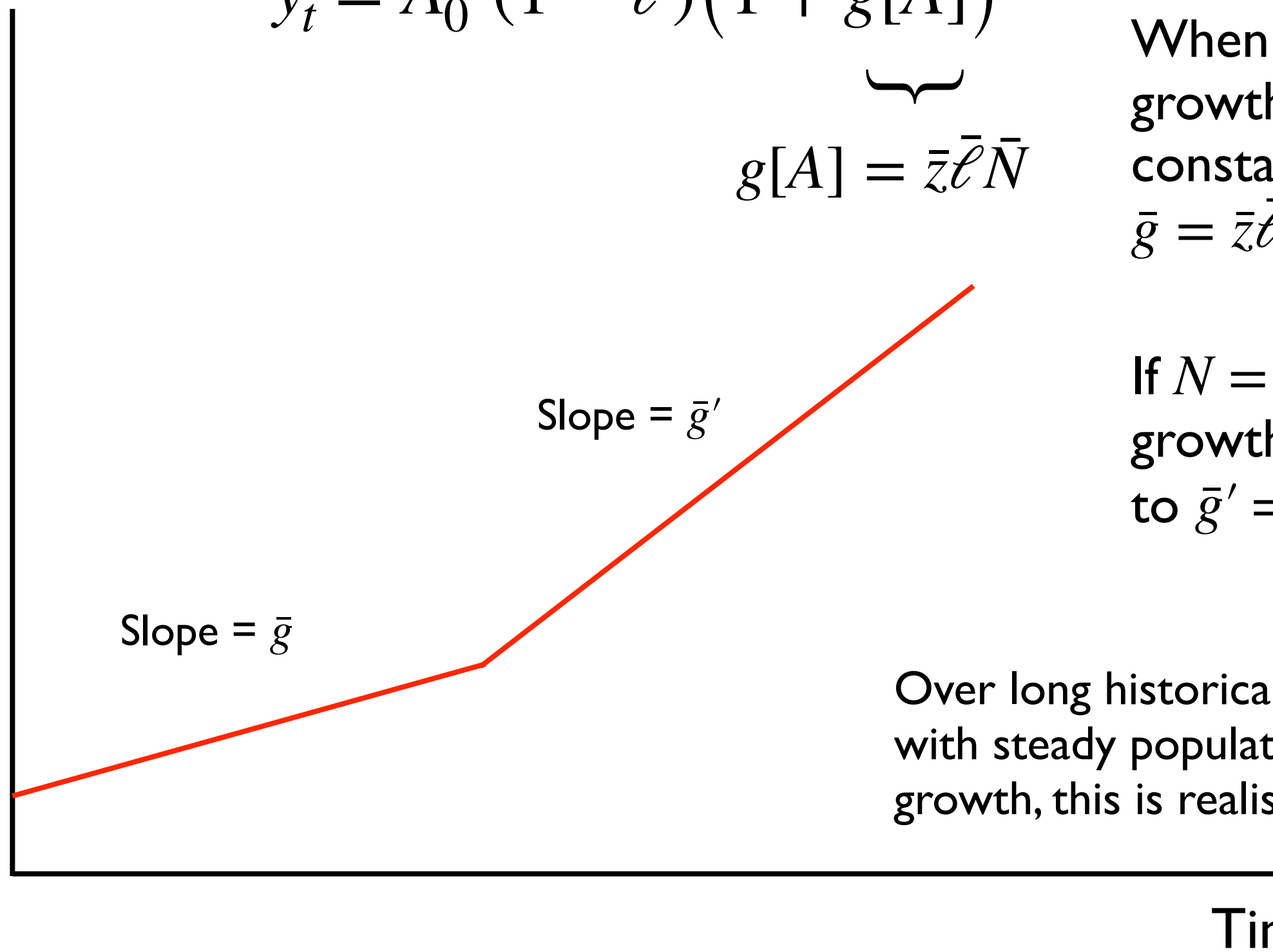
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Ratio scale: y_t

$$y_t = A_0 (1 - \ell) \underbrace{\left(1 + g[A]\right)^t}_{g[A] = \bar{z}\bar{\ell}\bar{N}}$$

When $N = \bar{N}$,
growth is
constant at
 $\bar{g} = \bar{z}\bar{\ell}\bar{N}$

If $N = \bar{N}'$, then
growth increases
to $\bar{g}' = \bar{z}\bar{\ell}\bar{N}'$

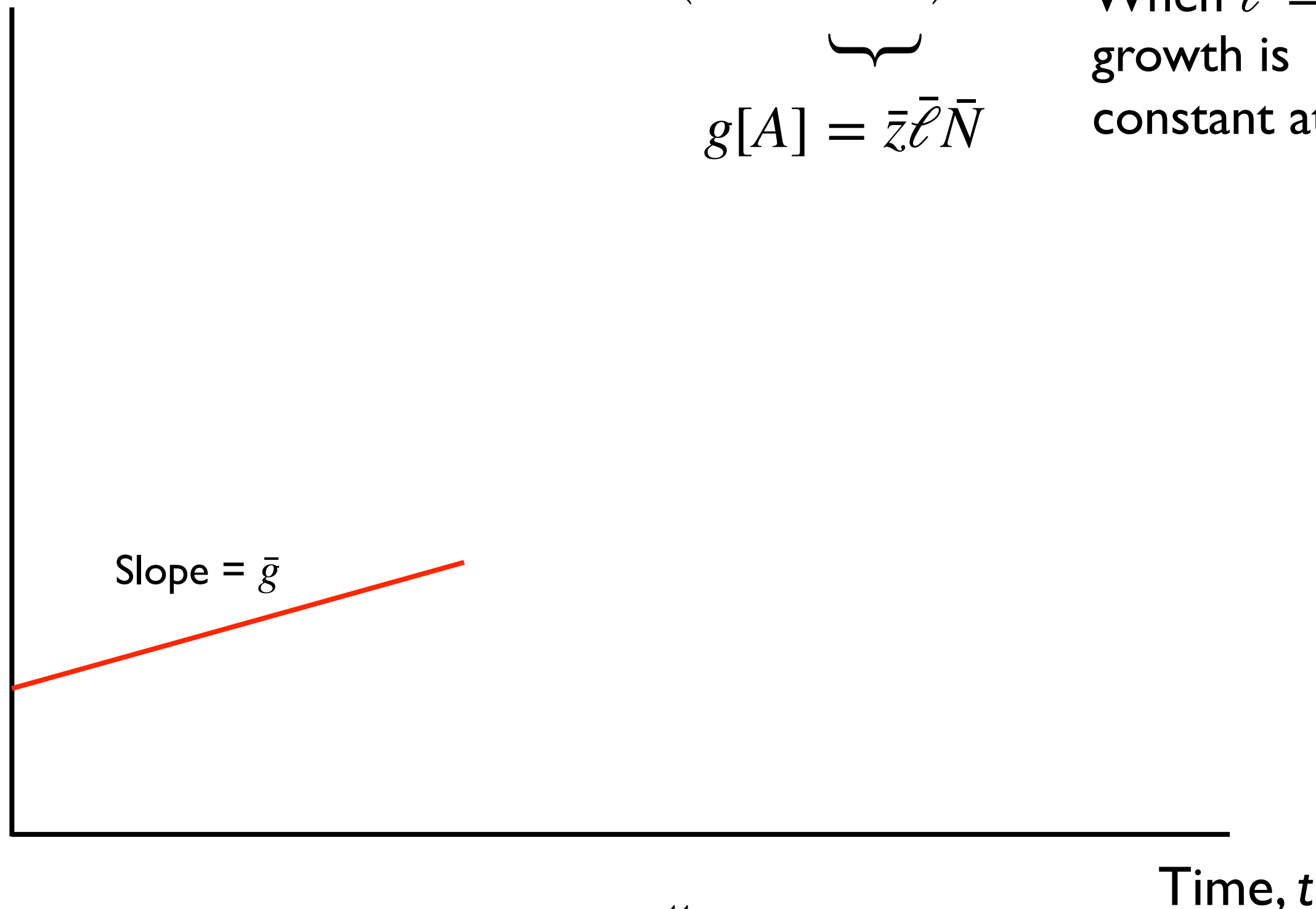


Suppose $\bar{\ell}$, the labor share of scientists increases

Ratio scale: y_t

$$y_t = A_0 (1 - \ell) \left(1 + \underbrace{g[A]}_{g[A] = \bar{z}\bar{\ell}\bar{N}} \right)^t$$

When $\ell = \bar{\ell}$,
growth is
constant at $\bar{g}$



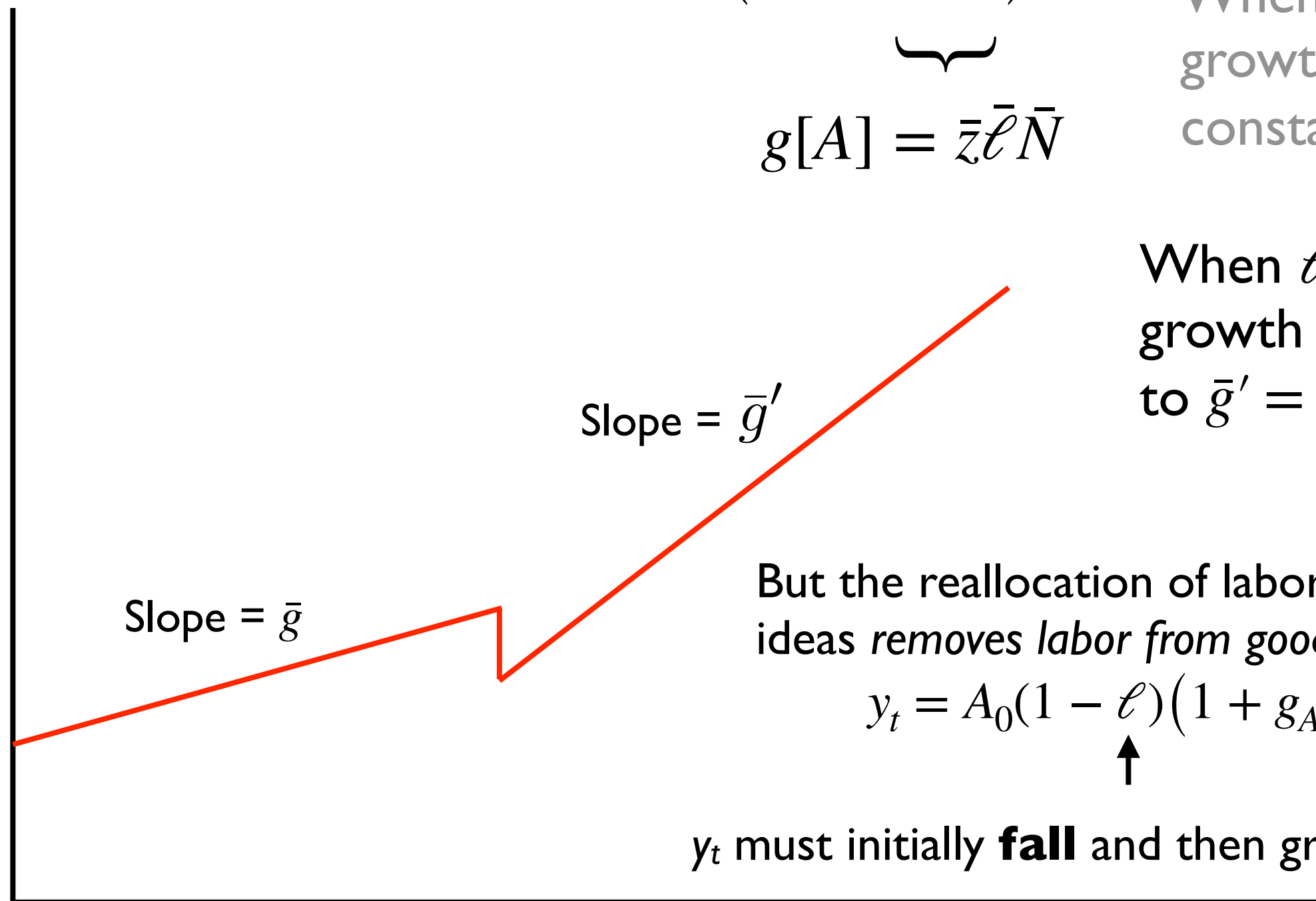
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Ratio scale: y_t

$$y_t = A_0 (1 - \ell) \underbrace{(1 + g[A])^t}_{g[A] = \bar{z}\bar{\ell}\bar{N}}$$

When $\ell = \bar{\ell}$,
growth is
constant at $\bar{g}$

When $\ell = \bar{\ell}' > \bar{\ell}$
growth increases
to $\bar{g}' = \bar{z}\bar{\ell}'\bar{N}$



Slope = $\bar{g}'$

Slope = $\bar{g}$

But the reallocation of labor toward
ideas *removes labor from goods production:*

$$y_t = A_0 (1 - \ell) (1 + g_A)^t$$



y_t must initially **fall** and then grow faster

Time, t